INCH-POUND

MIL-M-52863A
6 September 1994
SUPERSEDING
MIL-M-52863
14 May 1975

MILITARY SPECIFICATION

MAT, LANDING; ALUMINUM: HEAVY DUTY, AND ANCILLARY ITEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a heavy-duty expedient airfield surfacing system consisting of landing mats and ancillary items.

1.2 Classification. The components, mats, and ancillary items, shall be of the following types as specified (see 6.2):

Landing mat types

I - Full panel
II - Female half panel
III - Repair panel

Ancillary items

1 Adapter, access.
2 Anchor attachment.
3 Edge anchor assembly.
4 Bar, locking.
5 Starting connector.
6 Turn adapter, female.
7 Turn adapter, male.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5680

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-55057	- Panels, Wood/Wood Based; Construction and Decorative.
FF-N-105	- Nails, Brads, Staples, and Spikes: Wire, Cut and Wrought.
PPP-B-601	- Boxes, Wood, Cleated Plywood.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, For Aluminum and Aluminum Alloys.
MIL-C-81346	- Compound, Deck Covering, Nonslip, Lightweight for Aluminum Alloy Landing Mats.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurements.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of Military Property.
MIL-STD-621	- Test Method for Pavement Subgrade, Subbase and Base Course Materials.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MIL-STD-889	- Dissimilar Metals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13220E5650	- Landing Mats Assembly.
13216E9236	- Edge Anchor Assembly.
13220E5630	- Mat, Landing, Full Mat, Heavy Duty.
13220E5631	- Mat, Landing, Half Mat, Heavy Duty.
13220E5633	- Bar, Locking.
13220E5634	- Connector, Starting.
13220E5635	- Adapter, Access.
13220E5636	- Turn Adapter, Male/I-Lock.
13220E5637	- Turn Adapter, Female/I-Lock.
13220E5639	- Anchor Attachment, I-Lock.
13220E5640	- Repair Panel Assembly, Heavy Duty.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from: USA BELVOIR RDE CTR, ATTN SATBE JBS, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

ALUMINUM ASSOCIATION

Handbook	- Aluminum Standards and Data.

(Application for copies should be addressed to: ALUMN ASSN, 900 19TH STRET NW, WASHINGTON DC 20006.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to: AMERCN SCTY OF MECHL ENGRS, 345 E 47TH STRET, NEW YORK, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to: AMERCN SCTY FOR TEST & MTRLS, 1916 RACE STRET, PHILADLEPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The landing mat and ancillary items shall be in accordance with TA13220E5650 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the mat. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The mats shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

encountered in any of the applicable operating and storage environments to which the mats may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the mats may be newly fabricated from recovered materials to the maximum extent practicable, provided the mats produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the mats.

3.4 Interchangeability. When tested as specified in 4.5.2.1, the landing mat shall assemble and be fully interchangeable with like mats.

3.5 Structural capability and reliability. When tested as specified in 4.5.2.2 the landing mat, when placed on a subgrade with a California Bearing Ratio (CBR) of 4, shall have a 0.90 probability of withstanding 1,000 coverages of a 50,000-pound wheel load that has a tire pressure of 250 psi and a tire footprint of 200 square inches.

3.6 Identification marking. The mats shall be identified in accordance with MIL-STD-130, and as specified herein.

3.7 Finishing.

3.7.1 Chemical conversion coating. The mat shall be cleaned and given a chemical conversion coating in accordance with MIL-C-5541, class 1A.

3.7.2 Nonskid wearing surface. After the mat has been given a chemical conversion coating, the upper surface of the mat shall be coated with nonskid coating conforming to MIL-C-81346, grade 1 or 2 as applicable, except that the color shall be color no. 34087 in accordance with FED-STD-595. The thickness of the nonskid wearing surface shall be not less than 5 and not more than 10 mils thick, and cover the entire wearing surface. Nonskid compound shall not be applied to the connector rails.

3.7.3 Lower surface of mat. The lower surface of the mat shall be coated as specified in 3.7.1, and no painting is required.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

3.7.4 Ancillary items. Ancillary items, except for locking bars, shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Locking bars shall be anodized in accordance with MIL-A-8625, type I or II, class 1.

3.8 Metal stamping. Each mat and ancillary item, except for locking bars, shall be metal stamped on an exposed position near one end of component with the National Stock Number shown on the drawing. Numerals shall be .50-inch where space permits. On small components, metal stamping shall be as large as practicable.

3.9 Type I landing mat. Type I landing mat shall be as shown on 13220E5630, and as specified herein. The mat shall weigh not more than 116 pounds ±3 pounds.

3.10 Type II landing mat. Type II landing mats shall be as shown on 13220E5631, and as specified herein. The mat shall weigh not more than 60 pounds ±2 pounds.

3.11 Type III landing mat. Type III landing mats shall be as shown on 13220E5640, and as specified herein. The mat shall weigh not more than 141 pounds ±2 pounds.

3.12 Ancillary items. Ancillary items shall be as shown on the following drawings, and as specified herein:

Item	Drawing	Nomenclature
Ancillary 1	13220E5635	Adapter, Access.
Ancillary 2	13220E5639	Anchor Attachment.
Ancillary 3	13216E9236	Edge Anchor Assembly.
Ancillary 4	13220E5633	Bar, Locking.
Ancillary 5	13220E5634	Starting Connector.
Ancillary 6	13220E5637	Turn Adapter, Female.
Ancillary 7	13220E5636	Turn Adapter, Male.

3.13 Workmanship.

3.13.1 Metal fabrication. Metal used in the fabrication shall be free of kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Aluminum alloys shall be handled and stored in accordance with the procedure outlined in the Aluminum Association's Handbook, Aluminum Standards and Data. Alloys specified on the drawings shall not be permitted to come into contact with corrosive materials during storage. Cutting of aluminum alloys shall be accomplished by sawing, routing, or machining methods. Flame or arc-cutting, or shearing shall not be permitted.

3.13.2 Welding procedure. Prior to the production fabrication of any welding, the contractor shall prepare and submit to the

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

contracting officer for approval an isometric or perspective drawing showing the location of each joint and the joint welding procedures to cover all welding, including a general outline for the repair of base metal and welded joints. The procedures shall be qualified in accordance with ASME Boiler and Pressure Vessel Code, Section IX. All welds that are to be qualified shall be made on the same alloy, gage, and temper of materials, using the filler alloy specified on the drawings and the same procedure and positions as will be used in production. Weld procedures which have been qualified under other equivalent recognized codes may be considered qualified for the purpose of this specification, provided they are approved by the contracting officer. Any change in procedure during fabrication shall require requalification in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section IX.

3.13.3 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. Welds shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter.

3.13.4 Welders and welding operators. Before assigning any welder or welding operators to welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder or welding operators has passed qualification test as prescribed by the ASME Boiler and Pressure Vessel Code, Section IX, for the type of welding operation to be performed. Test specimens shall be made on the same alloy, gage, and temper of material with the same filler alloy and in the same position as will be used in production. Welders or welding operators who have been similarly qualified under other recognized codes may be considered qualified for the purposes of this specification, provided they are approved by the contracting officer. Operators of automatic welding equipment shall be considered welders.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, and unless disapproved by the Government, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Parts and components. Parts and components detailed on drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring inspection, the sampling plan and the basis for acceptance and rejection (see 6.5 and 6.6.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. First article mats and ancillary items shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article mats and ancillary items shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1.

4.4.3 Tests. The landing mats and listed ancillary items shall be tested as specified in 4.5.2.

4.5 Inspection procedure.

4.5.1 Examination. The mats and ancillary items shall be examined as specified herein for the following defects:

101. Any part of component not in accordance with the QAP requirements as shown on the drawings.
102. Materials not as specified (see 3.3).
103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (3.3.1.1).
105. Identification marking not as specified (see 3.6).
106. Finishing not as specified (see 3.7).
107. Nonskid compound not applied to the mat wearing surface as specified (see 3.7.2).
108. Connector rails not free from nonskid compound as specified (see 3.7.2).
109. Chemical conversion not applied to bottom of mat as specified (see 3.7.3).
110. Ancillary items not anodized as specified (see 3.7.4).
111. Metal stamping not as specified (see 3.8).
112. Weight of mats not as specified (see 3.9, 3.10 and 3.11).
113. Metal fabrication not as specified (see 3.13.1).
114. Nonconformance to approved welding procedure (see 3.13.2).
115. Welds not free from the specified defects (see 3.13.3).
116. Welders not certified as specified (see 3.17.4).
117. Workmanship not as specified (see 3.13).

4.5.2 Tests.

4.5.2.1 Interchangeability of landing mat system. Assemble nine full panels and one half panel in three rows as shown in figure 1 with locking bars. Three full panels and one half panel shall be randomly selected from each 750 panels and assembled as follows: Remove panels 2, 4, 6, and 9 and install the panels selected. Any binding of connectors, misalignment of mat or connector, or failure to assemble locking bars and mat, shall constitute failure of the test (see figure 1).

4.5.2.2 Structural and reliability test of the mat system. When specified (see 6.2), the landing mat system shall be tested as specified in 4.5.2.2.1 through 4.5.2.2.5. Failure of more than six of the 60 mats scheduled for traffic coverage (see figure 2) to survive all coverages shall constitute failure of the test. An individual mat failure is defined as any evidence

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

of separation of mats, fractures or deformation resulting in tire safety hazard or cracks exceeding 6 inches in length.

4.5.2.2.1 Preparation and testing of the subgrade. Construct a section of subgrade not less than 20 feet by 48 feet by 2 feet using a heavy clay soil. Place and compact the soil in 6-inch lifts. The CBR shall be determined in accordance with MIL-STD-621, and the CBR of each lift placed and of the subgrade after construction shall be not less than 3.5 and not more than 4.0. The CBR of the finished subgrade shall be taken at the surface and at depths of 6 and 12 inches at two locations. The two locations shall be 7 feet from and on opposite sides of the longitudinal centerline and one-third of the total distance from each end.

4.5.2.2.2 Installation of mats. Install type I and II mats on the subgrade as shown on figure 2. Type III mats will be used as replacement mats for the type I mats. Additional mats for approach around may be installed at both ends of the test section (see figure 2) to form a transition zone. The ends of the mats along the longitudinal edges of the surfaced area shall be anchored with the following anchoring system:

Anchor Attachment	13220E5639
Edge Anchor Assembly	13216E9236
Bar, Locking	13220E5633

4.5.2.2.3 Traffic distribution (coverages). Subject the installed mat to 1000 coverages of an aircraft tire inflated to a pressure of not less than 250 pounds per square inch (psi). The load applied to the tire shall not be less than 50,000 pounds, and the contract area shall be not more than 200 square inches. Apply the load at not more than 4 miles per hour (mph).

4.5.2.2.3.1 Coverage application. Coverages shall be applied in the 11 by 48 foot center of the 36 by 48 foot test section (see figure 2). Coverage shall be started at one side of the test lane and then back in the same path to complete two coverages. The tire shall then be shifted laterally the width of the tire print, and the procedure repeated a sufficient number of times to cover the width of the test lane. The interior 108 inches of the test lane shall then receive six additional coverages and the innermost 60 inches another two coverages, making a total of 10 coverages for the center 60 inches of the test lane. This cycle shall constitute 10 coverages. Subsequent coverages shall be applied by repeating this cycle as necessary to obtain the required number of coverages as determined by 4.5.2.2.4.

4.5.2.2.4 Equivalent coverages. The required equivalent coverages shall be calculated prior to starting the test, and at the completion of an equivalent of 500 coverages. The required coverages are calculated prior to the test using the CBR determined by 4.5.2.2.1 in the following equation:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

$$Log_{10}\ Equivalent\ Coverages = \frac{140.48}{\sqrt{\frac{6173}{CBR} - 64}} - 0.65$$

At the completion of an equivalent of 500 coverages, the remaining required coverages shall be calculated by using an average of all 12 CBR values taken as specified in 4.5.2.2.1 and 4.5.2.2.4 in the above equation. The value obtained must be divided by 2 to obtain the remaining coverages required. If the subgrade must be reconstructed, the coverages remaining shall be determined by using the CBR values obtained from the reconstructed subgrade to obtain coverages required to complete the test. The value obtained from the above equation shall be divided by 2.

4.5.2.2.5 CBR check during coverage application. At the conclusion of an equivalent of 500 coverages, the CBR shall be determined as specified in 4.5.2.2.1 at two locations within the test lane. Failure of the CBR to conform to the range of values specified in 4.5.2.2.1 shall be reason to stop the test and reconstruct the subgrade to the specified CBR. After failure of a mat during coverage application, the CBR of the subgrade directly beneath the failed mat shall be determined at the surface, and at depths of 6 and 12 inches. Failure of the CBR to conform to the range of the CBR values specified in 4.5.2.2.1 shall be reason to stop the test and reconstruct the subgrade to the specified CBR.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

118. Locking bars not furnished for each mat in bundle (see 5.3.1).
119. Mat bundle not constructed as specified on figure (see 5.3.1).
120. The mat bundle not packed in a crate as specified in accordance with figure 4 (see 5.3.1).
121. Starting connectors not packed and strapped as specified (see 5.3.2).
122. Access adapters not locked together, packed and strapped as specified (see 5.3.3).
123. Turn adapters, male I-lock, not packed in a box and strapped as specified (see 5.3.4).
124. Turn adapter, female I-lock, not packed in a box and strapped as specified (see 5.3.5).
125. Dunnage not meeting requirements of figure 8.
126. Edge anchor I-lock attachment not packed in a box and strapped as specified (see 5.3.7).
127. Edge anchors not packed in a box and strapped as specified (see 5.3.8).
128. Marking missing, illegible, incorrect or incomplete (see 5.4).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article for examination within the time frame specified (see 6.2) to prove prior to starting production packing, that the packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.7).

5.2 Preservation. No preservation shall be required.

5.3 Packing. (see 6.8).

5.3.1 Truss web mats. The truss web mats with locking bars (hereinafter referred to as "mats") shall be bundled and crated 18 mats per pack in accordance with figures 3 and 4. A suitable jib shall be used by the contractor to aid in preparing the bundle shown on figure 3, to assure exact and uniform positioning of the mats. After securing with two straps, the bundle can be moved and positioned at the crate assembly area. Thirty-six half

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

mats secured in pairs with the locking bar as well as 18 repair mats shall be assembled as described above.

5.3.2 Starting connectors. Starting connectors (see figure 5A) shall be packed flat, 4 per layer with every other connector inverted, 3 layers per box, in a snug-fitting box approximately 9 feet, .063 inches in length, 11.938 inches in width and 4.50 inches in depth (inside dimensions) (see 6.9 and figure 7). Box shall be strapped around the circumference (ends and sides), and then strapped girthwise at each end plus 3 straps between the end straps, equally spaced (total of 6 straps). Strapping shall conform to requirements of ASTM D 3953, type 1, zinc-coated, .75 by .025 inches, and ASTM D 4675.

5.3.3 Access adapters. Access adapters (see figure 5B) shall be locked together in groups of 7 and packed 7 per layer, 4 layers per box, in a snug-fitting box approximately 9 feet, .125 inches in length, 12.50 inches in width and 6.125 inches in depth (inside dimensions). Box shall conform to requirements of figure 7 and be strapped as described in 5.3.2 and figure 7 (see 6.9).

5.3.4 Turn adapter, I-lock, male. Seventy male I-locks along with 70 locking bars (see figures 5C and 5E), shall be packed in a snug-fitting box conforming to the requirements of PPP-B-601, overseas type, style I (with skids). Three girthwise straps shall be used. A box to contain the above quantities of components, arranged in accordance with the configuration shown in figure 9, will have inside dimensions approximating 24 inches in length, 23.875 inches in width, 8.625 inches in depth. The skids, as well as the load components, are to be located parallel with the 23.875 inch dimension. Box shall be constructed with the filler panel parallel with the skids. Use filler cleats in the panels perpendicular to the skids. Three straps located parallel with the skids and conforming to ASTM D 3953, type 1, zinc-coated, .75 by .025 inches, and ASTM D 4675.

5.3.5 Turn adapter, I-lock, female. One hundred and twelve female I-locks along with 111 locking bars (see figures 5D and 5E) shall be packed and strapped in a box as required in 5.3.4. A box to contain the above quantities of components, arranged in accordance with the configuration shown in figure 8, will have inside dimensions approximating 24.438 inches in length, 23.875 inches in width, and 10.563 inches in depth (see 6.9).

5.3.6 Locking bars. Locking bars (see figure 5E) are not packed separately, but with the item with which they are to be used (see 5.3.1, 5.3.4, and 5.3.5).

5.3.7 Edge anchor I-lock attachment. Two hundred and eighty-eight edge anchor I-lock attachments shall be packed in a box as required in 5.3.4. A box to contain 288 of this component, arranged in accordance with the configuration shown in figure 10, will have inside dimensions approximating 30.063 inches in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

length, 26.75 inches in width, and 8.50 inches in depth (see 6.9.1). Skids are to run parallel with the width dimension.

5.3.8 Edge anchor. One hundred and forty-four edge anchors are to be packed in the arrangement and box construction shown in figure 11. The box shall conform to PPP-B-601, overseas type, style I. The arrangement of the anchors calls for a row of 8 on top of a row of 10. The following row of 10 shall be so arranged that the eye end of the anchor in the bottom row will be directly under the helix end of the anchor in the third tier, etc. It has been found that the final 16th tier of 8 loaded into the box may fit best if all anchors are laid in one direction. When fully loaded, the box will contain 8 tiers of 10 anchors each, and 8 tiers of 8 anchors each (see 6.9).

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature than may be helpful but is not mandatory.)

6.1 Intended use. The landing mats and ancillary items, when assembled, comprise an expedient surfacing system which is capable of supporting the operation of the F-111 and C130 type aircraft. The system is capable of sustaining the impact and braking forces of aircraft which have single wheel loads up to 50,000 pounds.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the first article model, and types of the mat and ancillary items required (see 3.2).
c. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some or all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
d. When the structural test is required and whether or not the Government will conduct the structural test (see 4.5.2.2), the contracting officer should specify the time frame for conducting the test.
e. Time frame required for submission of the first article (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Equivalent sorties. The 1000 coverages applied to the mat during the coverage test is equivalent to 5,260 sorties. A sortie is one take off and one landing.

6.5 Quality Assurance Provisions (QAP). The contracting officers should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Definitions.

6.6.1 Quality Assurance Provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to pack and mark the landing mats and accessories in accordance with this specification (see 5.1).

6.8 Level of packing (see 5.3). Only a single method of packing is suitable and adequate for this commodity. For purposes of the packing level, marking level A shall be used. Level C packing is not applicable to this item as the shipping container will also be used in the field as a storage container. As such it is susceptible to exposure to all environmental conditions in addition to rough handling. A lesser level of pack would require a repack prior to troop issue.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

6.9. Weight and cube of component packs.

Component Pack & Quantity	Weight (lbs) Net	Gross	Cube
a. Truss web mat - 18*	2088	2180	59.7
b. Starting connectors - 12	384	457	7.7
c. Access adapters - 28	630	707	9.2
d. Turn adapters, I-lock, male - 70*	315	356	4.9
e. Turn adapters, I-lock, female - 112*	414	456	5.8
f. Edge anchor, I-lock - 288	474	525	6.5
g. Edge anchor - 144	576	655	13.2

* Packed with locking bars.

6.9.1 Box dimensions. Box dimensions given in section 5 were based on actual packing of components of previous acquisition. As such, succeeding acquisitions, while still conforming to specification requirements, could have slightly different dimensions when packed as specified in section 5. Therefore, a first article pack should be made before boxes are obtained for the entire acquisition quantity.

6.10 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
a. 3.13	DI-H-24029A	Certification, Welding Qualification
b. 3.13.2	DI-MISC-80875	Welding Procedures
c. 3.13.4	DI-MISC-80876	Welding Procedure Qualification Test Report

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL),

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.11 Subject term (keyword) listing.

Airfield landing mat
Airfield runway
Airfield surfacing system
Airfield taxiway

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

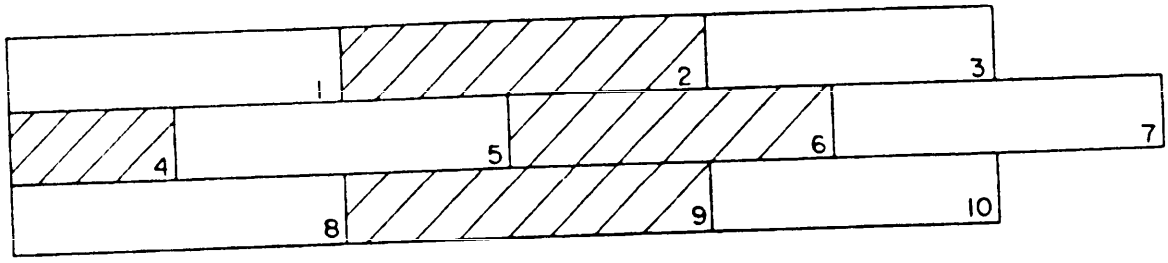
Custodians:
Army - ME
Navy - YD1
Air Force - 84

Review activities:
Army - CE
Navy - AS, MC

Preparing activity:
Army - ME

Project 5680-0197

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



(1) REMOVE PANELS 2, 4 (HALF PANEL), 6, AND 9.
(2) INSTALL SELECTED PANELS IN THESE VACATED POSITIONS AND CHECK FOR FIT.

STATIONARY PANELS

PANELS TO BE REMOVED

FIGURE 1. INTERCHANGEABILITY CHECK

X-2942

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

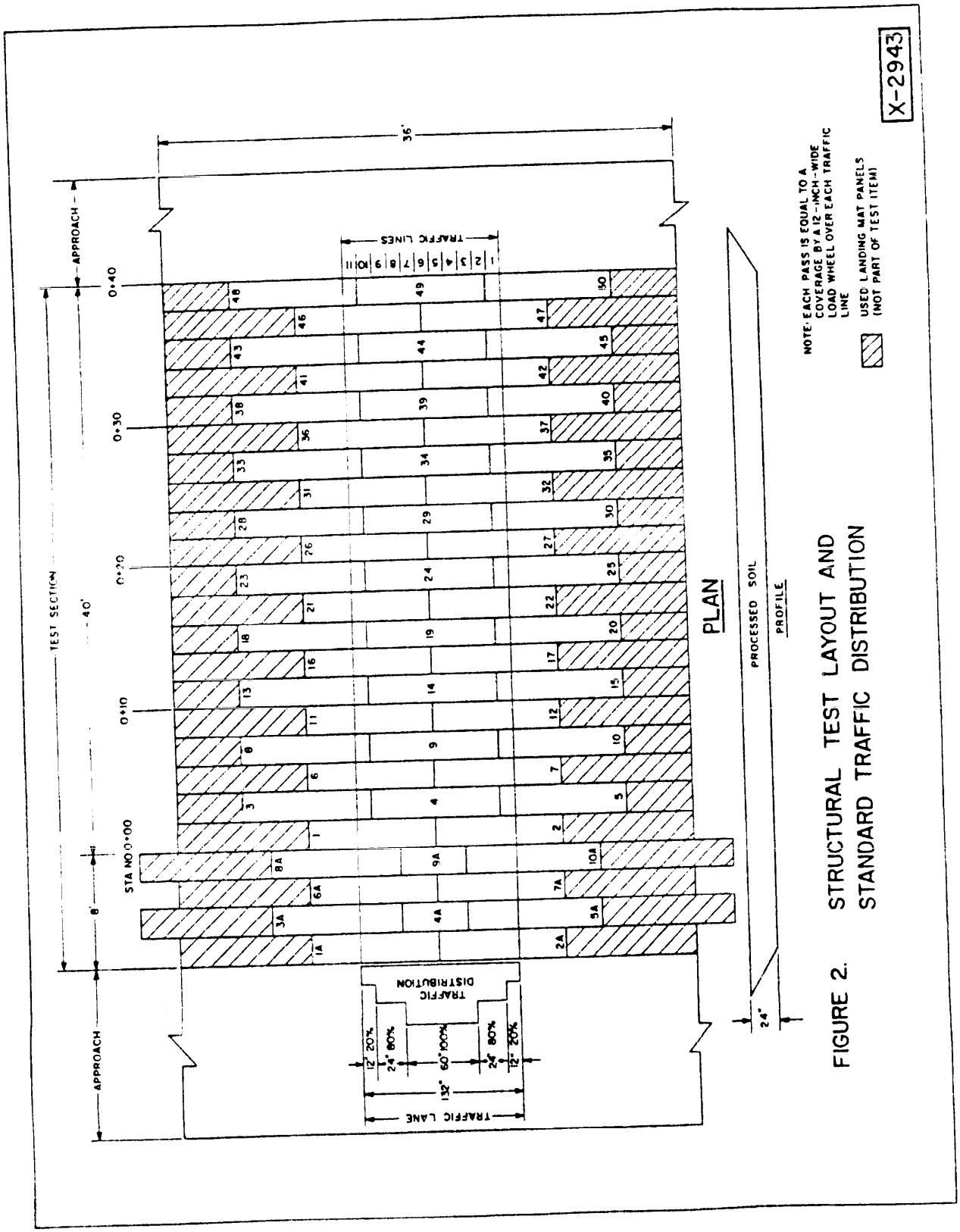
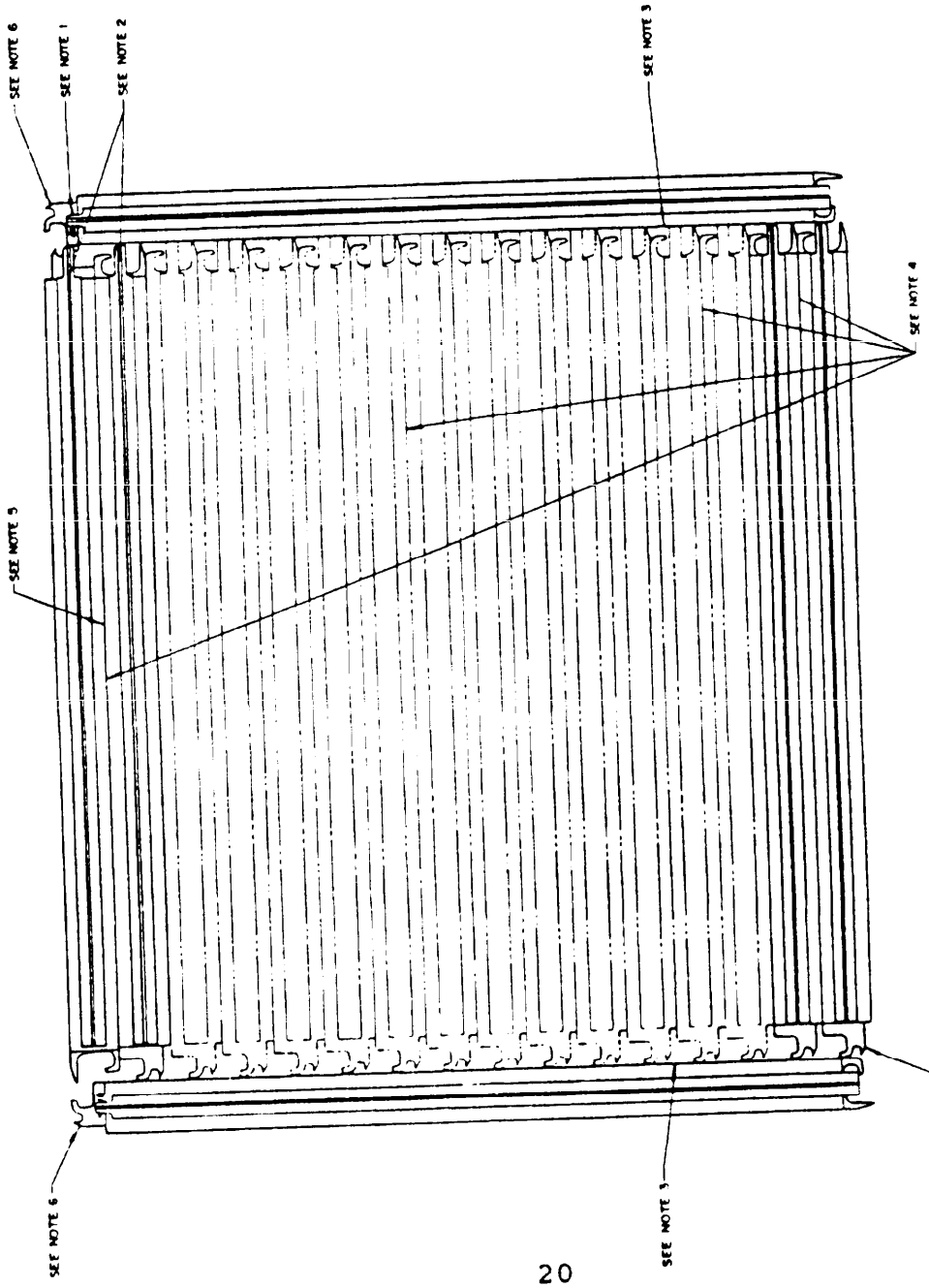


FIGURE 2. STRUCTURAL TEST LAYOUT AND STANDARD TRAFFIC DISTRIBUTION

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

NOTES:
1. THE VERTICAL LOCKING BARS ARE TO BE HELD IN POSITION PRIOR TO GRATING THE BUNDLE WITH A 1 IN. NO 12 WOOD SCREW INSERTED THRU THE HOLE IN THE LOCKING BAR
2. LOCKING BARS ATTACHED TO ALL MATS
3. TOP SIDE OF MAT TO INSIDE OF BUNDLE
4. TOP SIDE OF MAT UP
5. TOP SIDE OF LAST MAT DOWN
6. MALE CONNECTOR AT TOP
7. MALE CONNECTOR ALL ON ONE SIDE EXCEPT FOR TOP PANEL.
8. 18 FULL MATS, 36 HALF MATS OR 18 PREPAIR MATS POSITIONED IN THIS CONFIGURATION PRIOR TO STRAPPING WITH TWO GIRTHWISE STRAPS, LOCATED APPROXIMATELY 1/4 DISTANCE IN FROM EACH END AS SHOWN. STRAPPING TO CONFORM TO ASTM D3953, AND ASTM D4675, TYPE I, ZINC COATED, .75 X 0.029 INCHES. THIS FIGURE ALSO APPLIES TO 18 REPAIR MATS OR 36 HALF MATS
9. ALL DIMENSIONS ARE IN INCHES



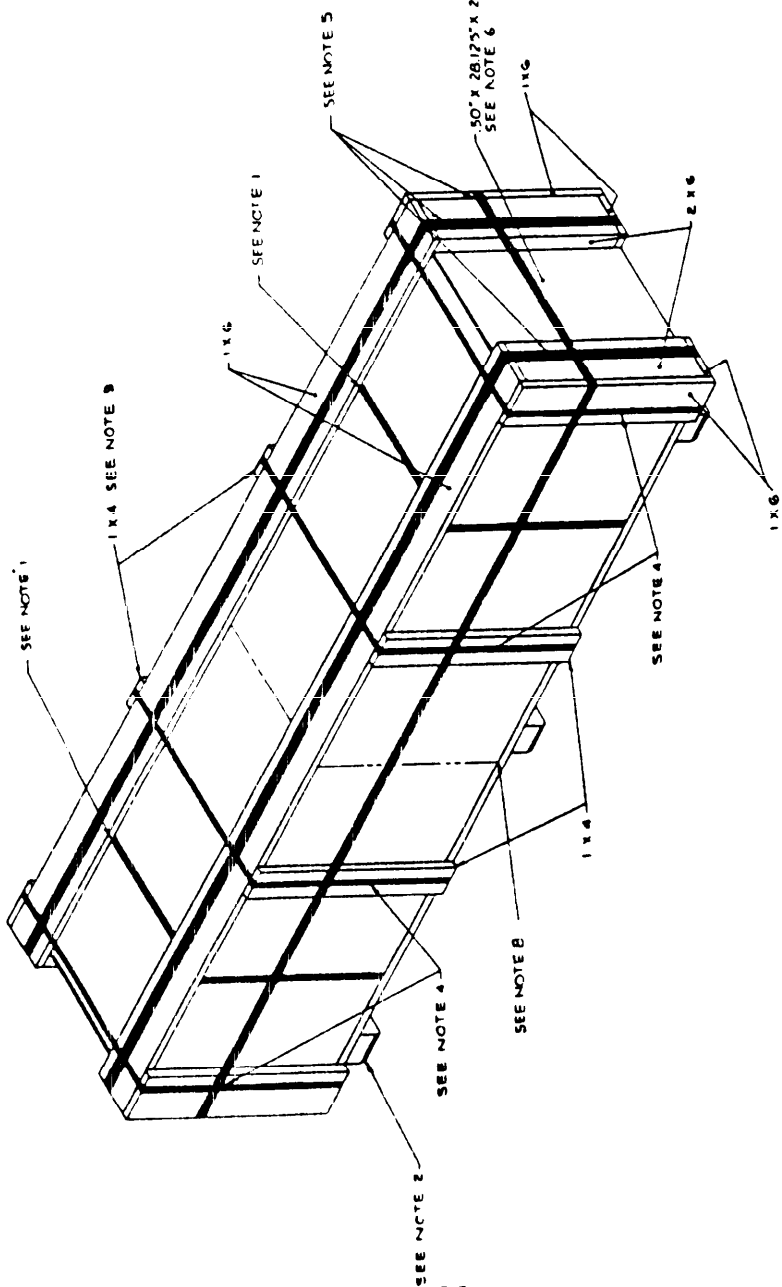
X-2963A

FIGURE 3.
PREPARATION OF MAT BUNDLE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

NOTES

1. BUNDLE STRAPPING BEFORE CRATING
2. NOM 3X4 SKIDS CHAMFERED ON CRATE SIDES SKIDS SECURED TO 1X6 MEMBERS WITH NAILS CONFORMING TO FF-N-105, TYPE II, STYLE 18, 2.50" L(MIN) NAILED THRU CRATE MEMBER INTO SKID, 3 NAILS PER CRATE MEMBER PER SKID. SKIDS TO BE NOTCHED FOR TWO LONGITUDINAL STRAPS, 1.50 INCHES X .25 INCH, STARTING 2 INCHES FROM END. SKIDS TO BE MOUNTED TO BOTTOM 6 INCHES IN FROM ENDS AND IN CENTER.
3. ALL NAILS, EXCEPT FOR SECURING SKIDS, TO CONFORM TO FF-N-105, TYPE II, STYLE 7, 8 OR 9, CEMENT COATED, 7d MINIMUM.
4. STRAPPING APPLIED OVER BATTENS CONFORMING TO ASTM D3953 AND ASTM D4675, TYPE I, ZINC COATED, .75 X 0.025 INCHES
5. LENGTHWISE STRAPPING APPLIED AS SHOWN TO CONFORM TO QQ-S-781, TYPE I OR IV, CLASS B, 1-1/4 X 0.035
6. PLYWOOD ENDS TO CONFORM TO A-A-55057, TYPE C, TYPE 1, GRADE 3-4 OR TYPE A, C-D INTERIOR, STANDARD SHEATHING WITH EXTERIOR GLUE, 1/2 INCH THK. SIZE OF PLYWOOD WILL BE DEPENDANT ON FINISHED MAT TO INCLUDE THICKNESS OF WELDS
7. WOOD MEMBERS OF CRATE TO CONFORM TO MIL-STD-731, WOOD, GROUP II OR III, GROUP 2.
8. VIEW WHEN 36 HALF MATS ARE CRATED.
9. LENGTH OF 1X4 BATTENS WILL BE LESS THAN THAT OF THE 1X6 MEMBERS MAT BUNDLE WILL SAG IN CENTER DUE TO END WELDS



X-2964A

FIGURE 4.
CRATING DETAILS FOR MAT BUNDLE
APPLICABLE TO WHOLE MATS, HALF MATS AND REPAIR MATS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

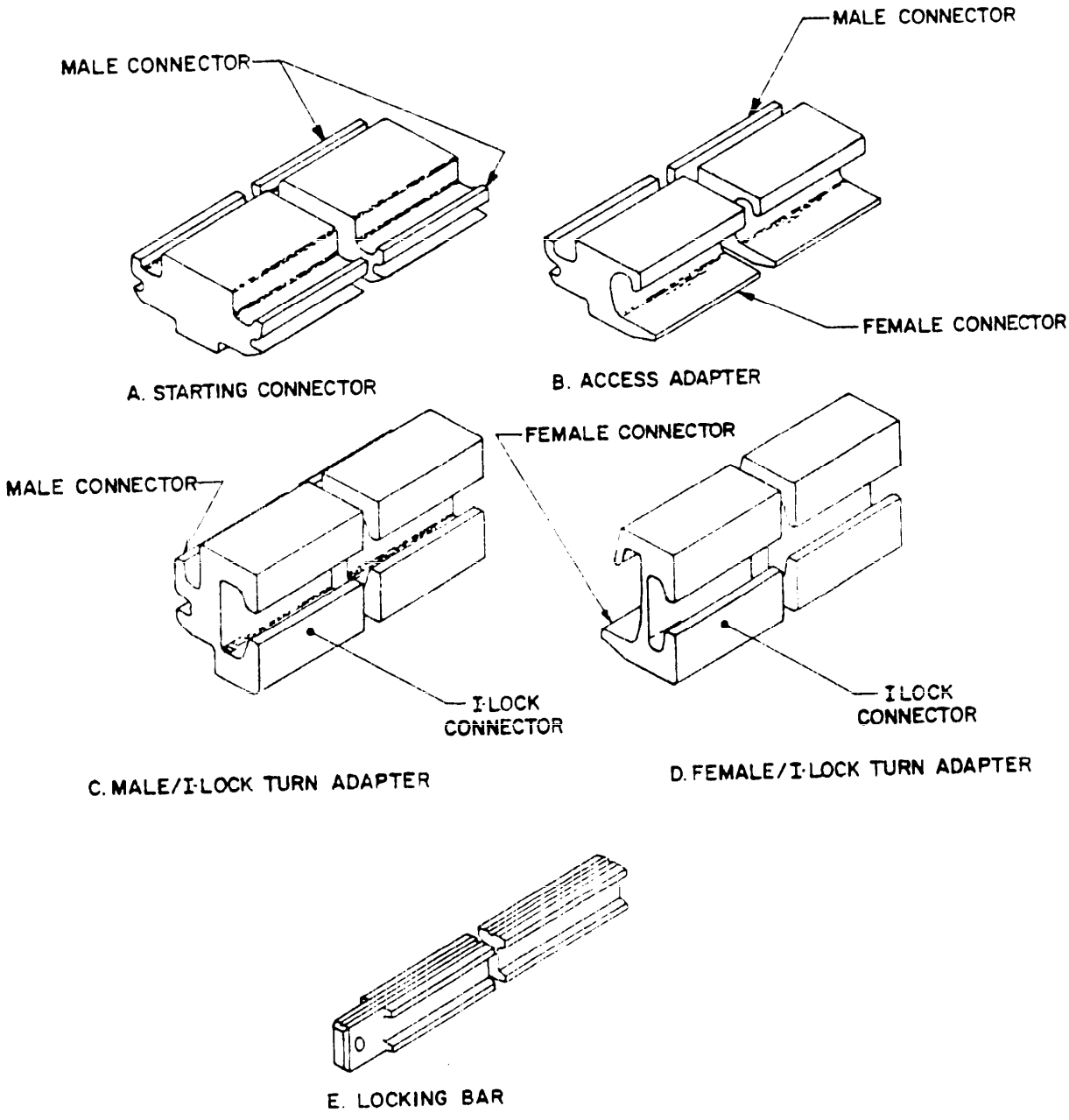
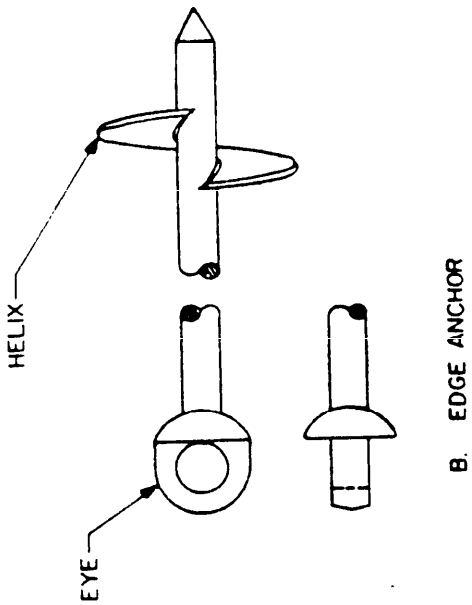


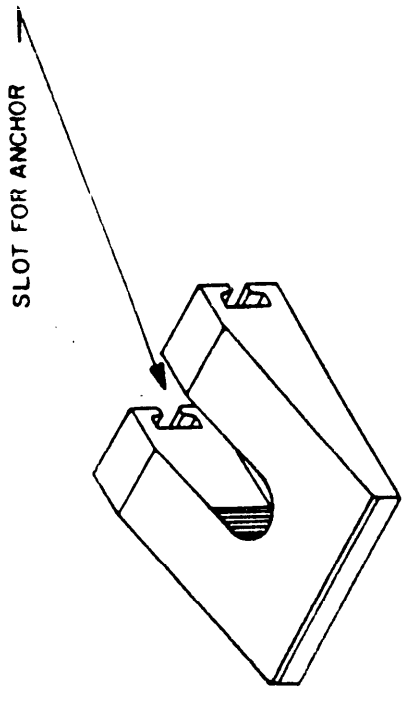
FIGURE 5.
VARIOUS ANCILLARY ITEMS

X-2965

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.



B. EDGE ANCHOR



A. I-LOCK ATTACHMENT

FIGURE 6.
EDGE ANCHOR AND ANCHOR I-LOCK ATTACHMENT

X-2966

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

NOTES:

1. SIDES, ENDS, TOP AND BOTTOM SHALL BE NOMINAL 1-INCH LUMBER CONFORMING TO REQUIREMENTS OF MIL-STD-731, WOOD, GROUP II OR III, CLASS 2.
2. SKIDS SHALL BE NOMINAL 3 X 4, CHAMFERED ON BOX SIDES. SKIDS SHALL CONFORM TO REQUIREMENTS OF NOTE 1 ABOVE, EXCEPT THAT CLASS 3 MATERIAL MAY BE USED. SEE FIG. 4 FOR SKID NAILING REQUIREMENTS.
3. INNER PORTION OF THE DOUBLE END BOX SHALL BE 3/4 INCH THICK PLYWOOD CONFORMING TO NOTE 6, FIG. 4.
4. NAILING AT THE ENDS SHALL BE INTO THE OUTER WOOD PORTION OF THE DOUBLE END. NAILS SHALL COMPLY WITH REQUIREMENTS OF NOTE 3, FIG. 4.

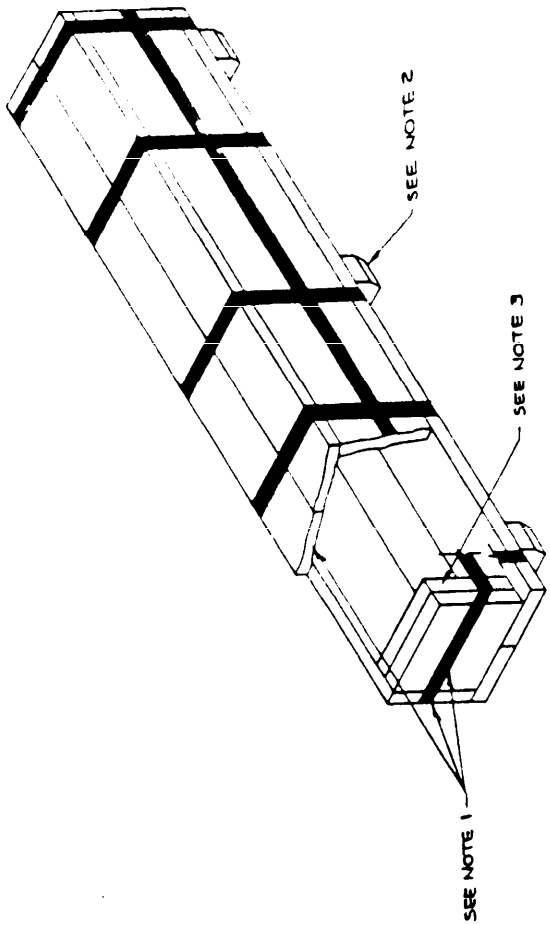


FIGURE 7.
CONSTRUCTION AND STRAPPING DETAILS, STARTING CONNECTORS AND ACCESS ADAPTERS

X-2967

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

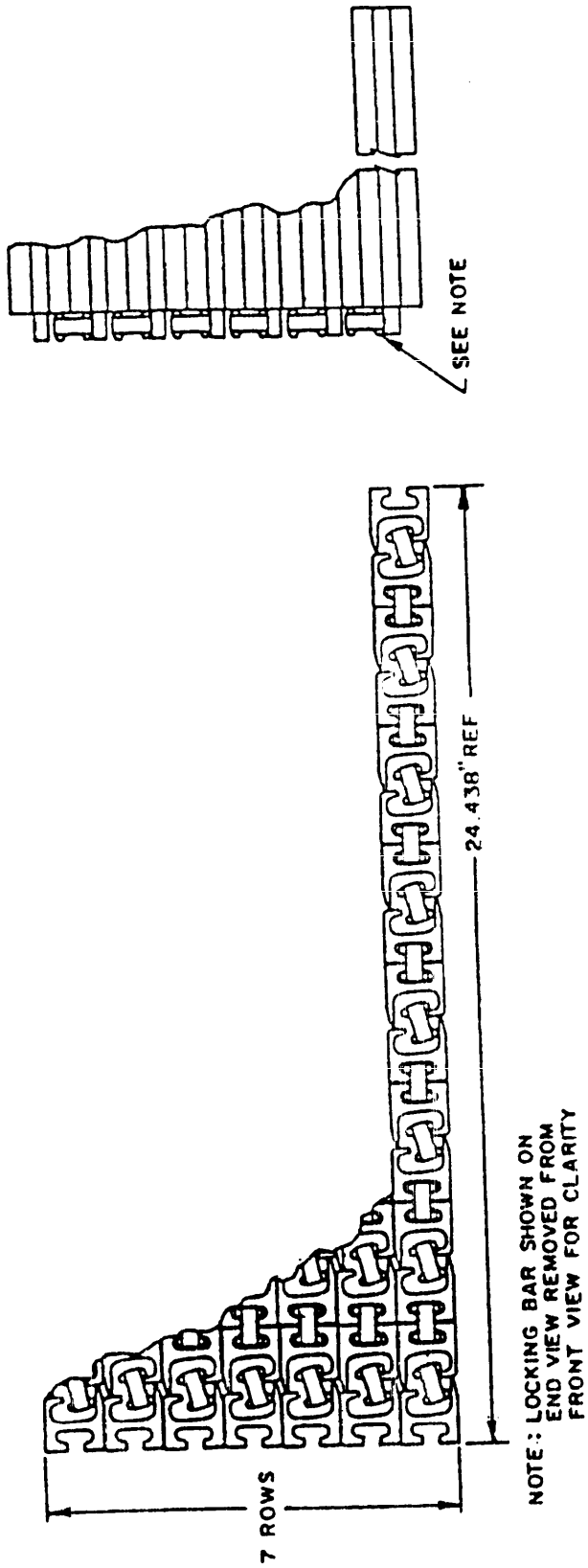


FIGURE 8. Packing configuration for turn adapter I-lock, female. 112 each along with 111 locking bars.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

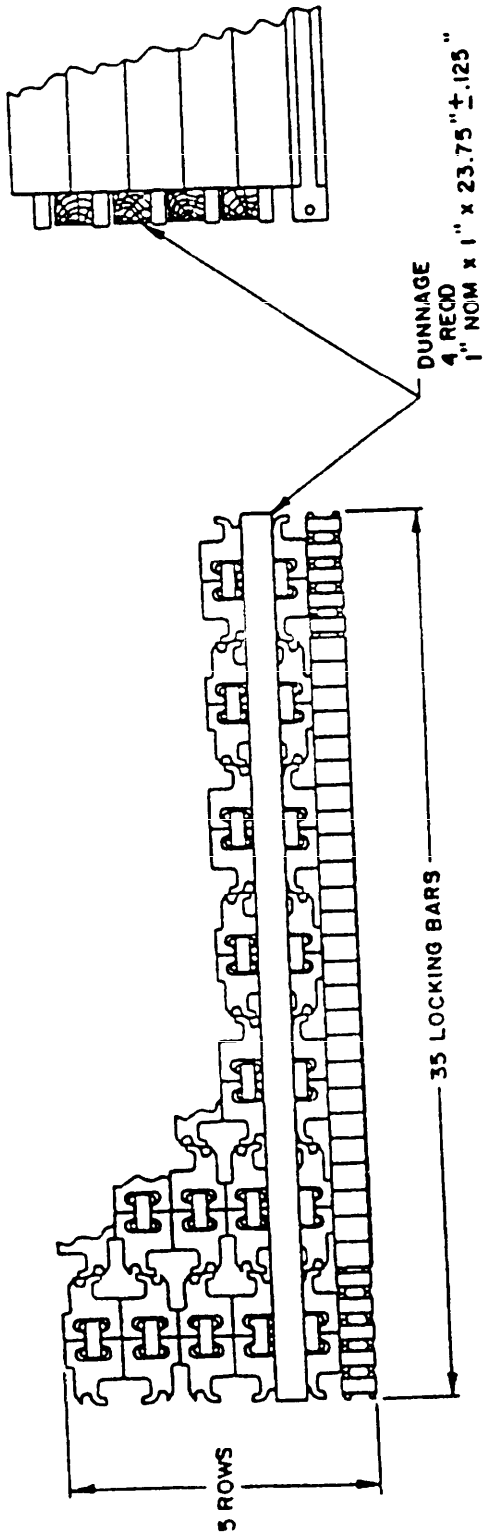
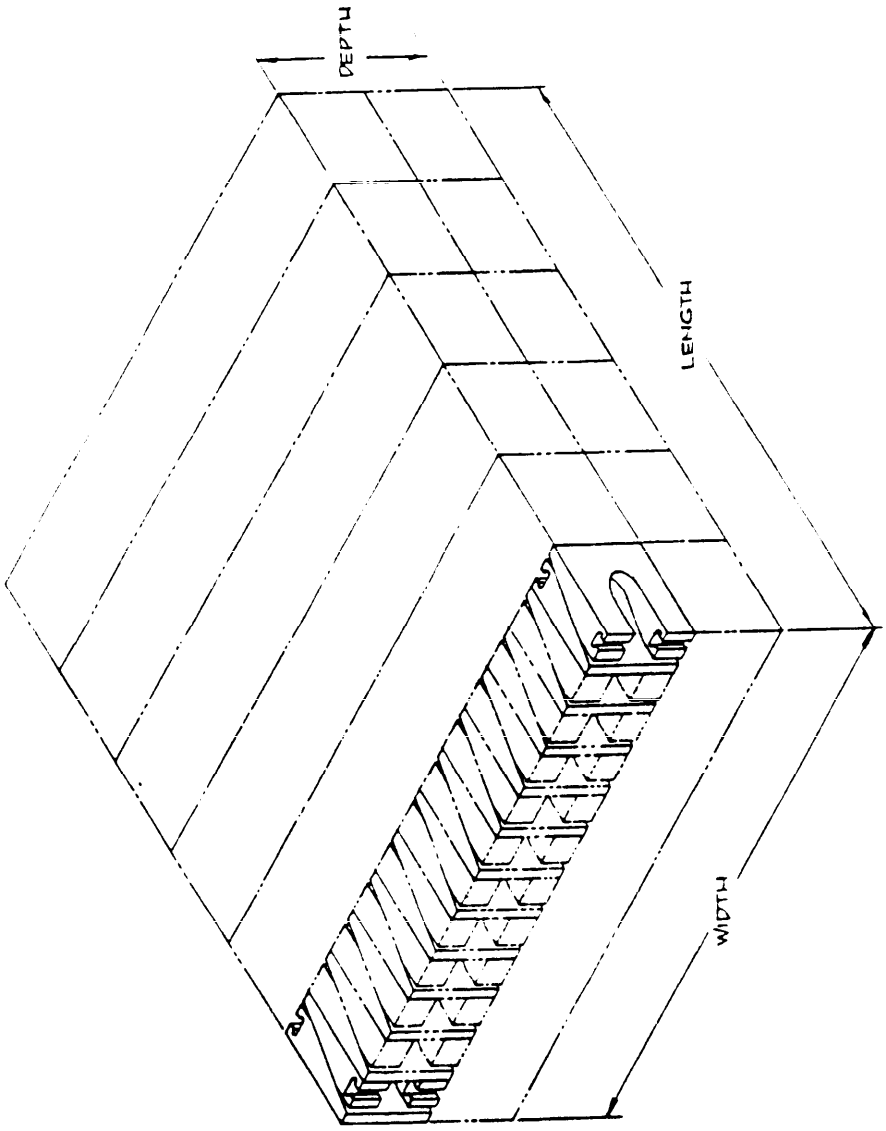


FIGURE 9. Packing configuration for turn adapter, I-lock, male, 70 each along with 70 locking bars.

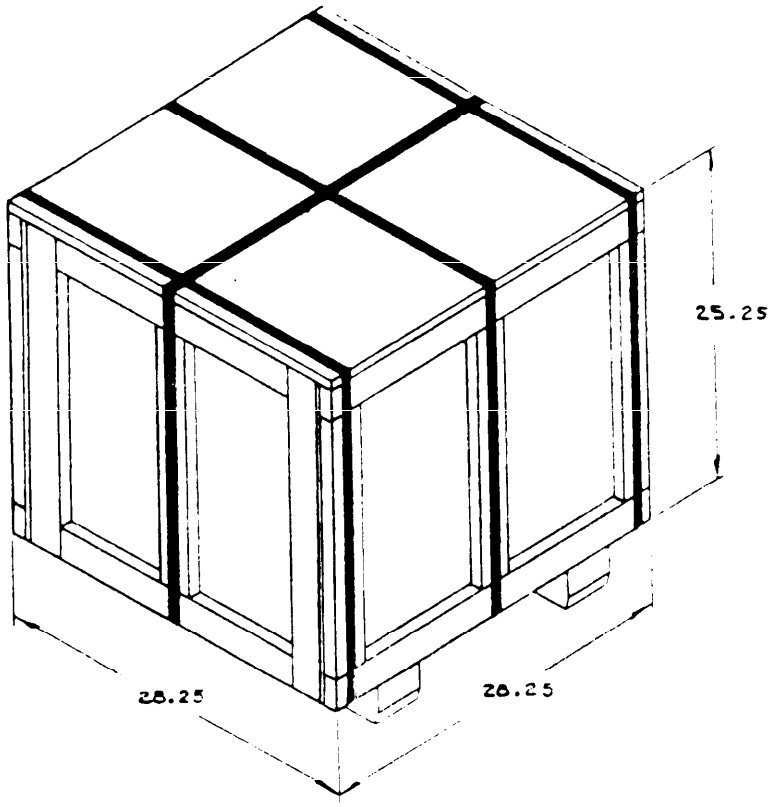
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



X-2970

FIGURE 10.
PACKING CONFIGURATION FOR
EDGE ANCHOR 1-LOCK ATTACHMENTS, 288 EA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. NOTCH SKIDS 1"X.25" IN CENTER.
2. STRAPPING TO CONFORM TO ASTM D3953 AND ASTM D4675, TYPE I, ZINC COATED, .75 X .025 INCHES.
3. SKIDS PARALLEL WITH SHORT PANEL.

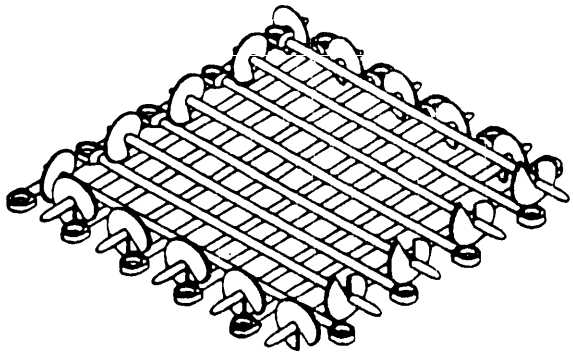


FIG. 11.

BOX DESIGN AND LOAD CONFIGURATION FOR EDGE ANCHORS.

X-2971A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-52863A	2. DOCUMENT DATE (YYMMDD) 940906

3. DOCUMENT TITLE
CUTTERS; PAPER, GUILLOTINE: FLOOR MODEL, ELECTRIC-MOTOR-DRIVEN

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) MY BELVOIR RDE CTR SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

...shall be those

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:10Z
Check the source to verify that this is the current version before use.
...conversion before use. This specification...

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.